LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

NCP Lab Rotation I

Schedule & Experiment Introduction

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on behalf of

BoSCA Lab



Your Tasks for this Lab Rotation & People BoSCA Lab

- Learn about the project & preparing a preregistration & report requirements
  - now, with me ☺
- Prepare the preregistration (**deadline 27.11.2026**)
- Pilot the experiment with Clara (**03.12.2026**)
- Collect data (**07.12.2026 – 18.12.2026)**
- Data Analysis with Simon **(15.01.2027)**
- Present the results **(27.01.2027)**
- Write a report (**deadline 15.03.2027)**

- Preregistrations and Reports should be sent to Simone!



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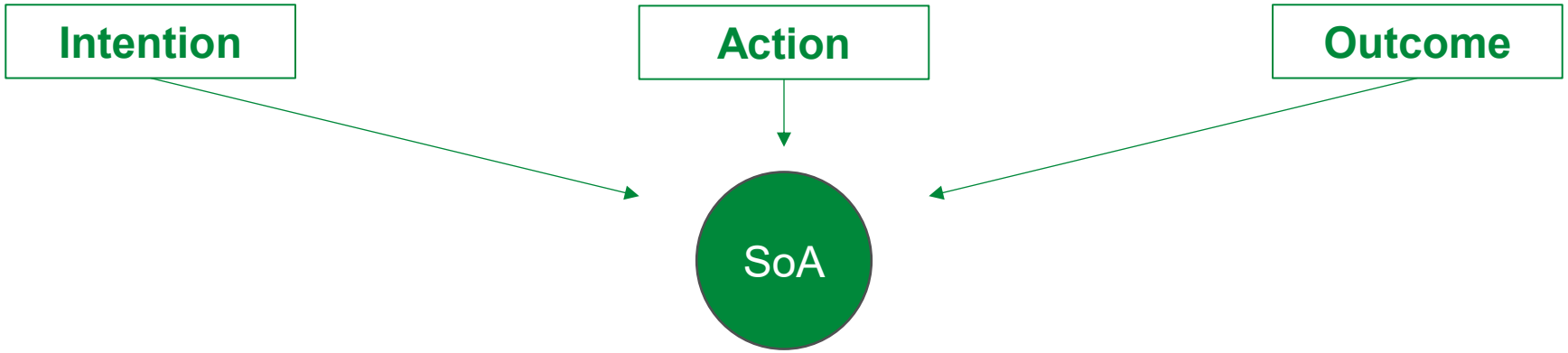
Today's Schedule

- Introduction of the experiment "In Control of Memory"
- BREAK
- Go over preregistration template
- Go over report requirements

In Control of Memory: The Role of Motor Control and Sense of Agency in Recognition Memory

Background

- Sense of agency (SoA) is *one's subjective* ***feeling of being in control*** *of their intentional actions, and their outcomes* (Haggard & Eitam, 2015).



In Control of Memory: The Role of Motor Control and Sense of Agency in Recognition Memory

Background

Lower-level	+	Higher-level
Sensorimotor control	e.g., Synofzik et al., 2008	**Action–outcome congruence** **Freedom of Choice**



Rafa Nadal





Rafa's Coach Toni

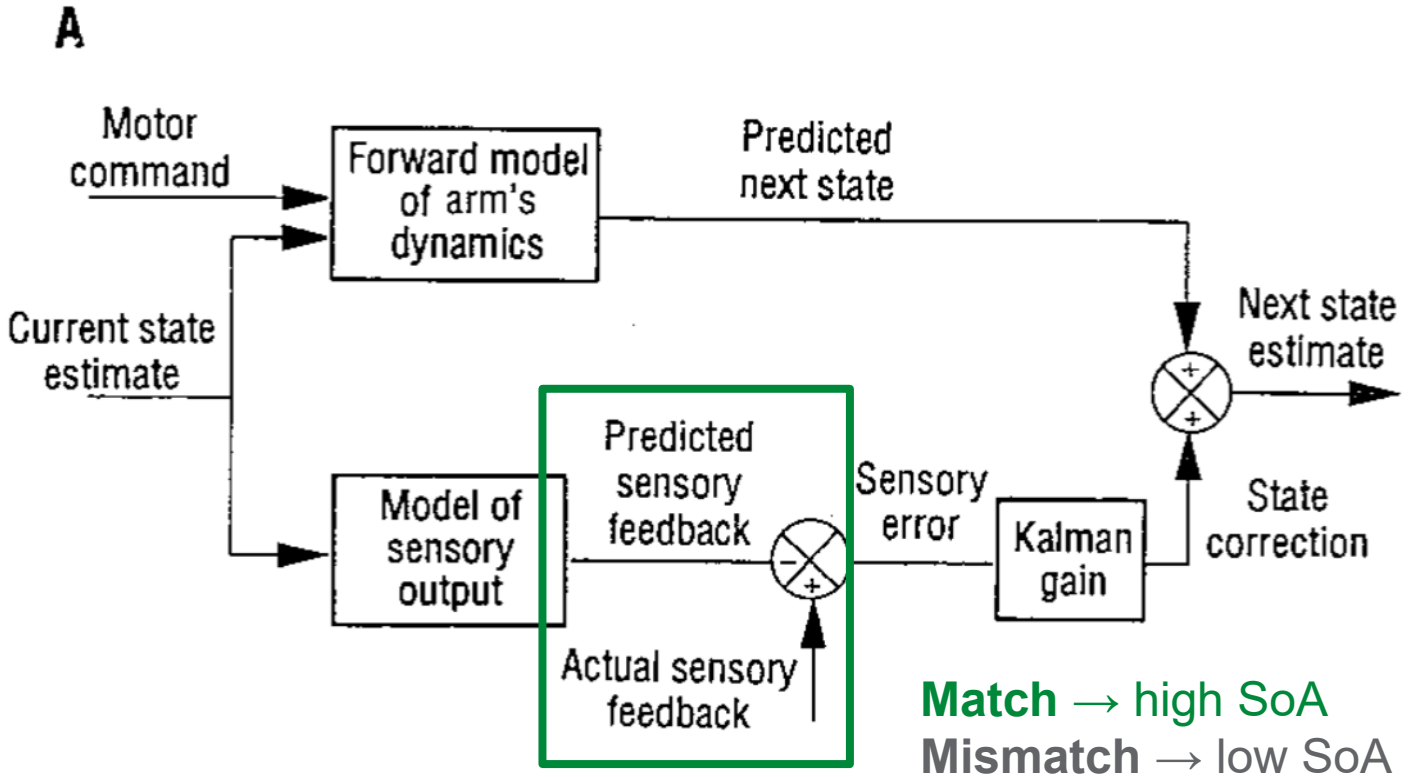


Referee

In Control of Memory: The Role of Motor Control and Sense of Agency in Recognition Memory

How is Sense of Agency Generated?

- Feedback Comparator Model (Wolpert et al., 1995)

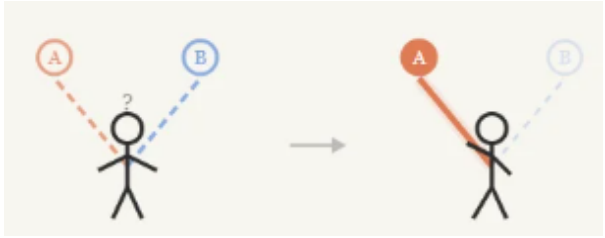


Match → high SoA
Mismatch → low SoA

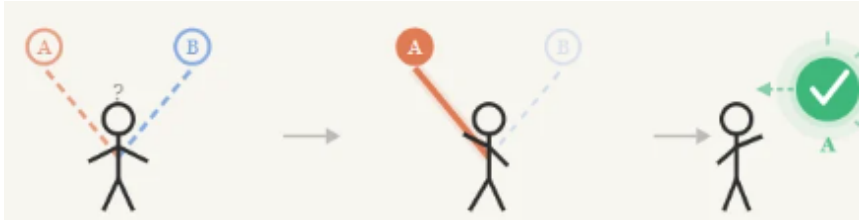
In Control of Memory: The Role of Motor Control and Sense of Agency in Recognition Memory

Consequences of SoA

- Sense of agency affects many other aspects of cognition:



Action selection (e.g., Wen & Haggard, 2018)



Feedback processing (e.g., Giersiepen et al., 2023; 2024)



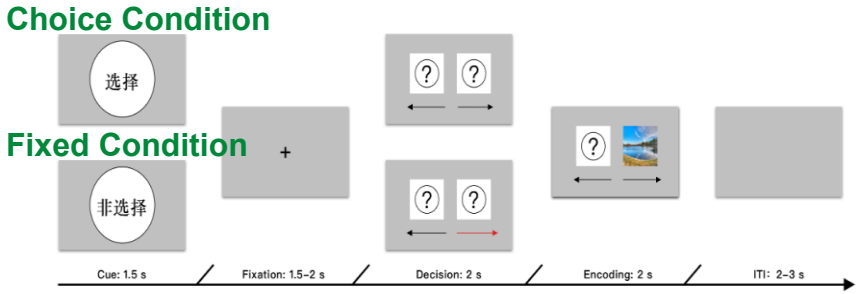
Memory (e.g., Cheng et al., 2023; Hon & Yeo, 2021; Ren et al., 2026; Schreiner et al., 2024; Tsuji & Imaizumi, 2022; Zhu et al., 2026)

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SoA & Memory

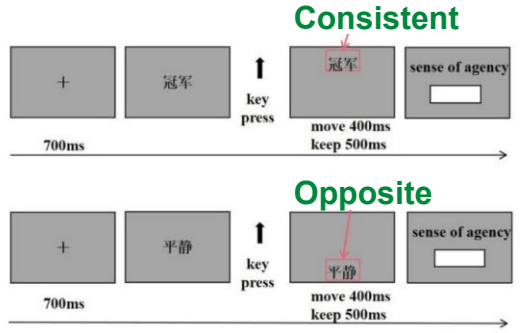
High SoA → better recognition memory

Lower-level		Higher-level
Sensorimotor control	e.g., Synofzik et al., 2008	**Action–outcome congruence** **Freedom of Choice**



Cheng et al., 2023

Outcome Monitoring



Zhu et al., 2026

In Control of Memory: The Role of Motor Control and Sense of Agency in Recognition Memory

SoA & Memory

High SoA → better recognition memory

Lower-level		Higher-level
Sensorimotor control	e.g., Synofzik et al., 2008	**Action–outcome congruence** **Freedom of Choice**

?

Discussion points:

- **Self-referential processing** (Hon & Yeo, 2021; Ren et al., 2026; Schreiner et al., 2024)
- **Feature binding** (Schreiner et al., 2024)
- **Reward circuitry** (Cheng et al., 2023)
- **Attentional prioritization** (Ren et al., 2026; Schreiner et al., 2024)

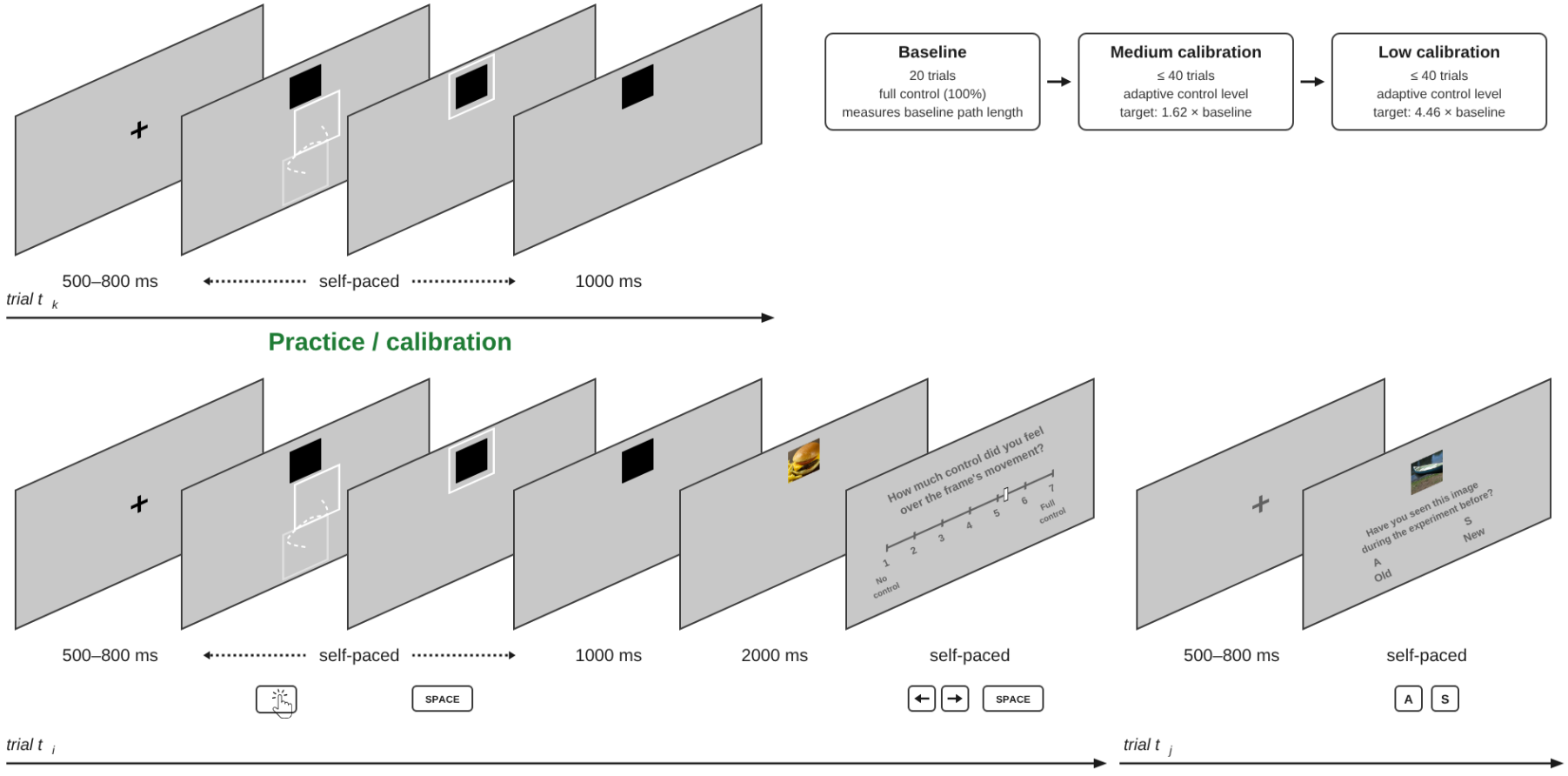
The Present Study

AIM

Investigate whether SoA, operationalized as motor control, also enhances recognition of incidentally encoded items.

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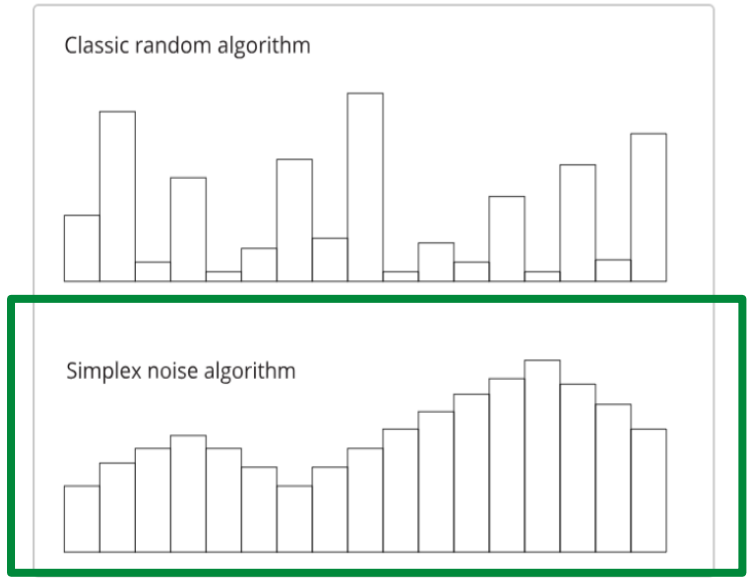
The Present Study – The Photo Shooting Task



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The Present Study – The Photo Shooting Task

perturbed position = cursor position + (1- control level) * Perlin noise * *k*



https://css-tricks.com/simulating-mouse-movement/

- 100% – full control
- *x*% – medium control
- *y*% – low control

within-subjects conditions

x and y are determined participant-specifically during calibration

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The Present Study – Hypotheses

H1 Felt control is highest under full, lower under medium and lowest under low control

Manipulation check

DV:

- Sense of agency rating (1–7, after every trial)

H2 Higher motor control leads to better recognition

Main hypothesis

DV:

- Higher accuracy (hits, i.e., “old” answers to old images)

H3 Higher motor control leads to faster recognition

Reaction times during recognition

DV:

- Faster recognition reaction times (hits only)

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The Present Study – Sample

Sample size

$N = 30$

≈ 4 recruited by each of you

Eligibility criteria

- Good command of English
- 18–45 years old
- Right-handed
- Normal or corrected-to-normal vision
- No color-vision deficiencies
- No known memory-related disorders

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The Present Study – Analyses per Hypothesis

H1 Felt control

Felt-control ratings

- Linear mixed model on the 180 ratings per person
- *rating ~ condition + (1 + condition | participant)*
- **Test:** F-test of condition, then F–M, F–L, M–L

H2 Recognition accuracy

Accuracy (hits)

- Logistic mixed model on the 180 old pictures per person
- *hit ~ condition + (1 + condition | participant) + (1 | picture)*
- **Test:** if significant, F–M, F–L, M–L as odds ratios; supported if Full > Low

H3 Recognition speed

Reaction time (hits only)

- Linear mixed model on log-transformed RTs
- *log(rt) ~ condition + (1 + condition | participant) + (1 | picture)*
- **Test:** F-test of condition, then F–M, F–L, M–L, supported if Full is faster than Low

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Preregistration

- A preregistration is a ***timestamped document*** describing your research questions, hypotheses, design and analysis plan ***before*** you conduct the study.

Why preregister?

- Be **transparent** about the research you are conducting
- Avoid **poor research practices** (e.g., HARKing, p-hacking)
- Curtail **publication and hindsight bias**
- Separate **confirmatory** from **exploratory** analyses
- Document **deviations** from the original plan openly

Where to preregister?

- Open Science Framework (OSF)
- ***AsPredicted***
- Leibniz Psychology (PreReg)
- Social Science Registry

Source: University of Mannheim Open Science Office, “Preregistration”, uni-mannheim.de/open-science/pre-registration

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Preregistration

Preregistration Template

1) Data Collection. Have any data been collected for this study already?

- Yes, we already collected the data.
- No, no data have been collected for this study yet.
- It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid **pre**registration nevertheless.

Note: 'Yes' is not an accepted answer.

2) Background. Briefly summarise the relevant research and explain how it motivates the study.

Example: Academic performance can decline during the summer holidays, particularly among children from disadvantaged backgrounds. Structured academic activities during this period may help maintain learning. This study examines whether offering a month-long academic summer programme can reduce this decline.

3) Research Question & Hypothesis. What's the main question being asked or hypothesis being tested in this study?

Example: A month-long academic summer program for disadvantaged kids will reduce the drop in academic performance that occurs during the summer.

4) Dependent Variables. Describe the key dependent variable(s) specifying how they will be measured.

Example: Simple average GPA across all courses during the first semester after the intervention.

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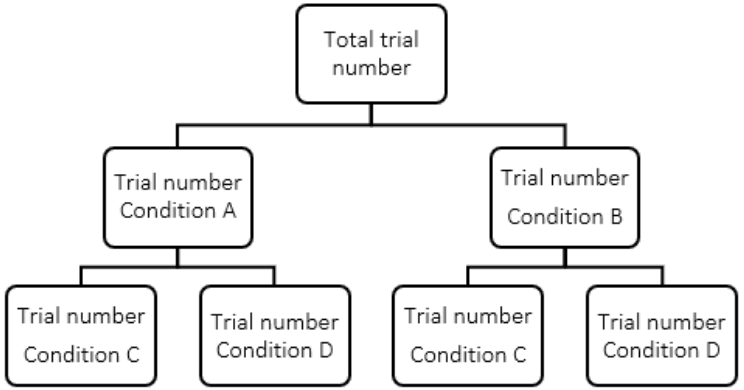
Preregistration

5) Experimental Design. Describe the task, experimental manipulations, and conditions. Specify whether participants complete all conditions or are assigned to separate groups. Report the number of blocks, trials per block, and total trials per participant, including the number of trials per condition. Include a figure illustrating the trial sequence.

Example 1: Two conditions: Offering summer program: yes vs no.

Example 2: 12 conditions in a mixed design lab study. Participants will be assigned to one of four conditions: math training, verbal training, memory task, or control (4 between-subject conditions). Each participant will complete a math test, a verbal test, and a memory test (3 within-subject conditions).





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Preregistration

6) Analyses. Specify the analyses and comparisons you will use to test each hypothesis.

Example. Linear regression predicting the simple average GPA in the semester after the intervention with a dummy variable indicating whether the participant was offered the summer program or not (intention-to-treat analysis). We will also conduct the same regression controlling for simple average GPA during the semester before the intervention, gender, & household income (an 8-point scale ranging from 1 = below $20,000 and 8 = above $150,000).

7) Predicted Results. Describe the results you expect based on your hypotheses.

Example. We predict that children offered the summer programme will have a higher average GPA in the first semester after the intervention than children who were not offered the programme, controlling for their GPA before the intervention.

8) Outliers and Exclusions. Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Example 1. We will compute the overall mean and standard deviation across all conditions, and exclude trials with a 2.5 SD above/below the mean.
Example 2: We will exclude participants who incorrectly answer at least 2 of our 3 attention check questions.
Example 3. We will exclude any participants who complete the survey in less than 30 seconds.

9) Sample Size. How many observations will be collected or what will determine sample size?
No need to justify decision, but be precise about exactly how the number will be determined.

Example: We will offer the program until 500 people have agreed to participate in it or until June 30, 2016 (whichever comes first).

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Preregistration

10) Other. Anything else you would like to pre-register?
(e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

Example: We will include a battery of questions for exploratory purposes, including life satisfaction, amount of videogame playing, and family activity. We will also provide an additional survey with 24 questions assessing achievement orientation. We will not report the results of those analyses for the project being pre-registered.
NOTE: If you leave this blank it will read: Nothing else to pre-register.

11) Name. Give a title for this AsPredicted pre-registration
Suggestion: use the name of the project, followed by study description.

Example: SUMMER PROGRAMS - GPA performance, Chicago, July 2018

12) Type of study.

- ○ Class project or assignment
- ○ Experiment
- ○ Survey
- ○ Observational/archival study
- ○ Other: _____

13) Data source

- ○ Prolific
- ○ MTurk
- ○ Cloud Research
- ○ University lab
- ○ Field experiment / RCT
- ○ Other: _____

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Requirements for the Report

Evaluation Criteria	Points	Comments
Realization of Given Tasks	of 5	
Attendance/Compliance with Deadlines	of 5	
Data Analysis	of 10	
Interpretation of Results	of 5	
Written Report: Abstract Content Writing Style	of 10 of 6 of 4	
Written Report: Introduction Content Writing Style	of 10 of 6 of 4	
Written Report: Methods Content Writing Style	of 10 of 6 of 4	
Written Report: Results Content Writing Style	of 10 of 6 of 4	
Written Report: Discussion Content Writing Style	of 10 of 6 of 4	
Written Report: Figures	of 10	
Written Report: APA Compliance	of 5	
Overall Student's Involvement/Motivation	of 10	

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Resources for Preregistration, Report, Experiment

https://github.com/elifgzn/NCP_LabRot_ws2627

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References

Cheng, S., Ding, Z., Chen, C., Sun, W., Jiang, T., Liu, X., & Zhang, M. (2023). The effect of choice on memory: The role of theta oscillations. *Psychophysiology, 60*(12), e14390. https://doi.org/10.1111/psyp.14390

Giersiepen, M., Schütz-Bosbach, S., & Kaiser, J. (2023). Freedom of choice boosts midfrontal theta power during affective feedback processing of goal-directed actions. *Biological Psychology, 183*, 108659. https://doi.org/10.1016/j.biopsycho.2023.108659

Giersiepen, M., Schütz-Bosbach, S., & Kaiser, J. (2024). My choice, my actions: Self-determination, not instrumental value of outcomes enhances outcome monitoring during learning. *Cerebral Cortex, 34*(8), bhae325. https://doi.org/10.1093/cercor/bhae325

Hon, N., & Yeo, N. (2021). Having a sense of agency can improve memory. *Psychonomic Bulletin & Review, 28*(3), 946–952. https://doi.org/10.3758/s13423-020-01849-x

Ren, Q., Herbelin, B., Meyer, N. H., Stampacchia, S., Schütz-Bosbach, S., & Blanke, O. (2026). Sense of agency strengthens memory for self-caused spatial information. *iScience*, 115382. https://doi.org/10.1016/j.isci.2026.115382

Schreiner, M. R., Feustel, S., & Kunde, W. (2024). Linking actions and memories: Probing the interplay of action-effect congruency, agency experience, and recognition memory. *Memory & Cognition*. https://doi.org/10.3758/s13421-024-01644-2

Tsuji, N., & Imaizumi, S. (2022). Sense of agency may not improve recollection and familiarity in recognition memory. *Scientific Reports, 12*(1), 21711. https://doi.org/10.1038/s41598-022-26210-1

Wen, W., & Haggard, P. (2018). Control changes the way we look at the world. *Journal of Cognitive Neuroscience, 30*(4), 603–619. https://doi.org/10.1162/jocn_a_01226

Zhu, R., Ma, H., Hommel, B., & Ma, K. (2026). Sense of agency promotes recognition memory: The role of self-referential processing. *International Journal of Psychophysiology, 229*, 113475. https://doi.org/10.1016/j.ijpsycho.2026.113475